EUROPEAN CENTRAL BANK

Metaphors and Inflation: A Study of European Central Bank (ECB) Communication

Do ECB Executive Members Use Conceptual Metaphors in Their Communications About Inflation? How Do They Use Them?

Data Science for Decision Making

Luis Francisco Alvarez, Sebastien Boxho, Mathieu Breier

July 2, 2024

Barcelona School of Economics

Acknowledgements

We would like to express our gratitude to Kristina Meier and the European Central Bank for generously providing the dataset and for the opportunity to collaborate on this project. Additionally, we extend our thanks to Hannes Mueller, Christian Brownlees, and Eliott Motte for their valuable assistance throughout the research process.

Contents

1 Introduction **4**
1.1 Context . . . 4
1.2 Motivation . . . 4
1.3 Related Literature . . . 5

2 Data **7**
2.1 Source . . . 7
2.2 Scraping . . . 7
2.3 Member Information . . . 8

3 Labeling **8**
3.1 Manual Metaphors Labeling . . . 8
3.2 Manual Categories Labeling . . . 8
3.3 GPT API . . . 9
3.4 Final Labels . . . 12

4 Metaphor Detection **13**
4.1 Regular Expressions (REGEX) . . . 13
4.2 Part of Speech (PoS) . . . 13
4.3 Neural Network . . . 14
4.3.1 Model Architecture . . . 14
4.4 Optimization . . . 15
4.4.1 Model Summary . . . 15
4.5 Pre-trained Large Language Model for Metaphor Detection . . . 15
4.5.1 Model Architecture . . . 16

5 Results **16**
5.1 Metaphors Identification . . . 16
5.2 Descriptive Analysis of Metaphors . . . 17
5.2.1 Analysis of Metaphor Usage by ECB Members . . . 18
5.2.2 Analysis of Metaphor Categories . . . 20

6 Conclusions **26**

A Appendix **28**

Abstract

This study systematically identifies and categorizes conceptual metaphors related to inflation in ECB board member communications, following the methodology outlined by Chunyu Hu and Zhi Chen in their paper, *"Inflation Metaphor in Contemporary American English"* [11]. Analyzing a corpus of ECB communications from 2005 to 2024, we identify and classify inflation-related metaphors using regular expressions (REGEX), part-of-speech (POS) tagging, and neural network models. Our findings reveal a consistent use of inflation metaphors across ECB executive board members, with notable variations in frequency and type depending on the individual and the time period. We leveraged the GPT API [19] to consistently label and categorize these metaphors, enabling robust performance analysis on the models and contributing to the emerging field of research in AI-driven text analysis.

Keywords: Conceptual Metaphors, Inflation, ECB Communications, Regular Expressions, Part-of-Speech Tagging, Neural Network Models, GPT API, AI-driven Text Analysis, Metaphor Classification

1 Introduction

1.1 Context

Inflation stands as a pivotal concern for economic policymakers and financial analysts globally, particularly exacerbated by the aftermath of the Covid-19 pandemic. The sudden resurgence of inflation has thrust it back into the forefront of economic discourse, sparking intense debates and policy considerations. Explaining inflation and justifying monetary policies in this context often demands intricate and nuanced communication. Among the European Central Bank (ECB) board members, metaphors emerge as indispensable linguistic tools. These metaphors play a critical role in clarifying an abstract concept such as inflation and its surrounding phenomena, making them more accessible to diverse audiences.

This study aims to systematically identify and categorize conceptual metaphors related to inflation in ECB board member communications, following the methodology outlined by Chunyu Hu and Zhi Chen in their paper, "Inflation Metaphor in Contemporary American English" [11]. In this study, 'inflation' is treated in a broad sense, encompassing metaphors related to disinflation and deflation as well. Leveraging advanced methodologies such as regular expressions (REGEX), part-of-speech (POS) tagging, and neural networks for enhanced word embeddings, we aim to dissect the metaphoric language employed. Additionally, the integration of the ChatGPT API will facilitate automated corpus labeling, enhancing efficiency and accuracy in metaphor identification.

By applying these analytical approaches, this study not only contributes to the understanding of how metaphors operate in economic discourse but also sheds light on their strategic use in communicating complex economic phenomena. The insights gained are expected to offer valuable implications for both academic research and practical policymaking strategies. More notably, this task can be insightful for further research into the way metaphors actually perform in a more qualitative way and their subsequent implications.

Finally, all of our working code and data files can be found in the Github repository [2].

1.2 Motivation

Metaphors, by their very nature, bridge the gap between abstract concepts and more tangible, easily understood elements from the real world. More specifically, conceptual metaphors are mental mappings from a (typically concrete) source domain to a (typically

abstract) target domain [11]. In our analysis case, inflation metaphors are implemented in order to facilitate the broader audience's understanding. However, they serve not only to clarify, but also to legitimize potentially unpopular monetary policy measures. This persuasive power is essential in garnering support and drawing attention to the complex dynamics of inflation. As in Den Hartog et al. 1997: [6] *"metaphors can serve not just to entertain but also to inspire and persuade"*. Therefore, it is noted by Chunyu Hu and Zhi Chen in their paper [11], "Inflation Metaphor in Contemporary American English", metaphors can significantly influence public perception and policy reception.

The main goal of this paper, as stated before, is to set up an automated method to uncover these metaphors, by employing a multi-faceted analytical approach. First, the integration of the ChatGPT API will facilitate our analysis, offering advanced natural language processing capabilities to identify and categorize metaphors while also serving as a proxy for human labeling and allowing for performance metrics calculation [28]. We will employ the GPT capabilities to replicate human labeling and later use these values for the evaluation of our models. Regex matching, Part of Speech tagging and Neural Network models will be applied for the identification of metaphorical language patterns, related to inflation, within ECB speeches.

By exploring the inflation metaphors used, this paper aims to answer the fundamental question: *do ECB board members actually use conceptual metaphors in their communications?* Our findings indicate that they do. The subsequent objective is to understand how these metaphors are used. This study will shed light on the rhetorical strategies in economic discourse, emphasizing the role of language in shaping economic narratives and the perception of monetary policies.

By identifying and classifying these metaphors, this research quantifies the frequency and types of metaphors used by ECB board members. Additionally, by leveraging large language models (LLMs) for metaphor identification, this paper facilitates future research on the practical impacts of metaphors, such as their role in building legitimacy for monetary policy measures, explaining abstract economic concepts, and serving as persuasive tools in ECB communications. This investigation thus contributes to a deeper understanding of the prevalence and types of metaphors in economic discourse and sets the stage for further exploration into their effectiveness.

1.3 Related Literature

The fundamental challenge of this study is to build an automated process to consistently identify the conceptual metaphors we have discussed, hence applying the Hu et al. [11]

findings over the ECB conferences. In this search, we will be using similar techniques to the ones implemented in the study "Political Metaphors in U.S. Governor Speeches" by Léo Picard and Dominik Stammbach [20]. This approach involves *"a deep- learning-based algorithm that scores word pairs (adjective-noun, subject-verb, or verb-object) by their metaphoricity. This metaphoricity score can be interpreted as the probability that humans would perceive a word pair as metaphorical."*.

To effectively identify metaphors within the ECB board member communications, we employ a model similar to the one outlined in the paper *Grasping the Finer Point: A Supervised Similarity Network for Metaphor Detection* by Marek Rei, Luana Bulat, Douwe Kiela, and Ekaterina Shutova [21]. This model employs a similarity neural network designed specifically for the task of metaphor detection. The approach combines the strengths of both neural networks and traditional (cosine) similarity measures, a common practice in natural language processing (NLP). It uses a supervised learning framework to train a network on a dataset of metaphorical and literal language examples. The network learns to identify metaphors by measuring the similarity between words in a context-sensitive manner, allowing it to grasp subtle differences in usage that signify metaphorical intent. Key components of the model include contextual embedding, which represents words within specific contexts to enable the model to differentiate metaphorical usage from literal usage, and similarity measurement, which calculates similarity scores between word pairs to help distinguish between these uses. By training on annotated datasets, the network predicts the likelihood of a given word being used metaphorically in a particular context.

This method enhances the accuracy of metaphor detection by incorporating a nuanced understanding of word meanings and their contextual relationships, rather than just comparing the "literal meaning" distance between the word pairs. Applying this model to ECB communications allows us to systematically identify and analyze the metaphors used, thereby contributing to a deeper understanding of the rhetorical strategies employed by board members.

The classification of these metaphors will be guided by the framework established by Hu and Chen [11], allowing for a structured understanding of how inflation is metaphorically conceptualized in the data corpus we shall be working on. This classification will reveal the underlying cognitive structures and rhetorical strategies employed by ECB board members to communicate their policies effectively.

2 Data

2.1 Source

Initially, we had a dataset of around 200 interviews provided by the ECB to work with. However, this data had some partitioning issues, an unstructured division between questions and answers, and several missing values. To address these problems, we decided to develop a web scraper to retrieve all the transcripts of interest directly from the Central Bank's website. The code can be found in the notebook: *00_ECB_Scrapper* and it parses the central bank's site using the Selenium package [24]. Instead of using the original dataset provided with the challenge, we will use our scraped data, which encompasses the entire original corpus and includes additional valuable information and observations.

2.2 Scraping

This section was instrumental in accessing additional information from each document, such as the date, the speaker, and the questioner. We were also able to, more accurately, partition questions and answers directly from the source, rather than implementing a preprocessing pipeline to extract this information from the text, as we had initially planned. This method not only reduces the likelihood of errors but also results in a more structured dataset to exploit for our task.

The resulting data corpus counts a total of 519 interviews spanning from 2005 to early 2024. In Figure1, we display the monthly number of interviews and the average European inflation rate, for the final 7 years of our period range and identify some key events, such as the Covid-19 pandemic and the outbreak of the Ukrainian-Russian war.

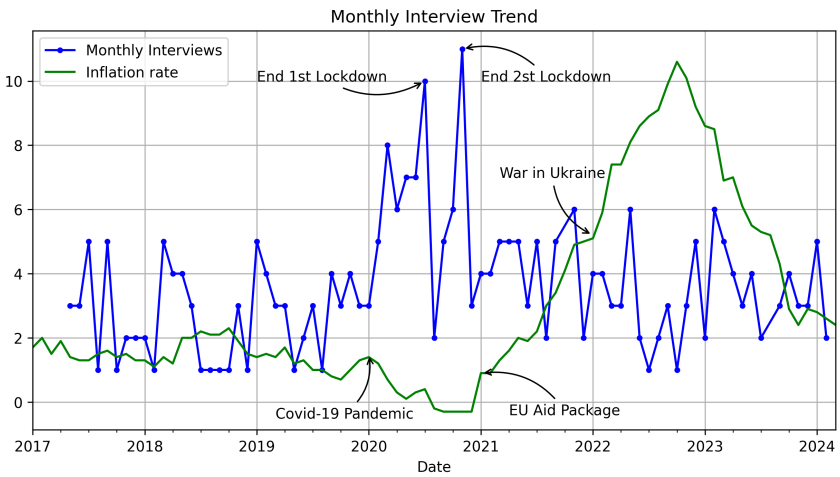

Figure 1: Monthly Interviews Trend and European Inflation Rate (2017 to 2024)

2.3 Member Information

The final step in our information gathering process for this thesis involves collecting individual data from the board members. We selected several features for all the speakers:

- Country of origin
- Gender
- Year of birth
- Period of service on the board

All information, were obtained directly from the ECB website [8]. Given that we are dealing with 19 different ECB members who have been part of the executive board, we opted to manually scrape the necessary information for each individual case. This additional data will prove valuable for subsequent quantitative analysis of the metaphors identified.

The corresponding code for this section is available in the notebook titled *"01_Speakers.ipynb"*.

3 Labeling

3.1 Manual Metaphors Labeling

Initially, we did not have labeled data, as our dataset consisted solely of scraped transcripts from ECB communications. Therefore, it was essential to label the data to train a supervised learning model capable of predicting metaphors.

Our approach involved manually labeling a subset of 68 interviews to identify the inflation-related metaphors in each one. We adhered to the methodology outlined in the Hu et al. [11] paper to ensure a consistent and rigorous labeling process. These manually labeled transcripts serve as our 'true labels', in order to compare with the GPT's output later.

3.2 Manual Categories Labeling

The other task at hand was to classify the identified metaphors into their respective categories. In 85% of the sentences containing the word 'inflation,' we found that the word was used as part of a metaphor, which leaves an average of 7.25 metaphors per interview.

We categorized our metaphors into the following 10 types:

Categories	
Machine	Liquid
Disease	Animal
Plant	Sports
Warfare	Other
Fire	Orientation

Table 1: Metaphor Categories

Some examples of the types of metaphors that can be found in our corpus data:

- **Inflation as warfare:** "[...] we allowed *inflation to become entrenched* in the economy" & "This monetary tightening, which we need to do to *kill inflation* [...]" & "[...] policies based around the *combating of inflation* [...]"
- **Inflation as machine:** "[...] restrictive policy has helped *cool down inflation*"
- **Inflation as an animal:** "*Taming inflation* in the early 1980s" & "[...] *feeding inflation*."
- **Inflation as fire:** "[...] he is sure to *fuel inflation* in the German economy [...]"

3.3 GPT API

The GPT-4 API's capability to grasp and classify metaphors is supported by findings in the paper "Does GPT-3 Grasp Metaphors? Identifying Metaphor Mappings with Generative Language Models" by Wachowiak and Gromann [28]. This study illustrates GPT-3's ability to understand and identify metaphorical language with a high degree of accuracy, providing a strong foundation for its use in our analysis of ECB communications. It demonstrates that generative language models like GPT-3, and subsequently GPT-4, possess a nuanced understanding of language, enabling them to detect and categorize metaphors effectively. Additionally, the robustness of GPT-4 in natural language understanding has been evidenced by various studies and applications, further validating its reliability for our purpose [16].

To ensure the accuracy of its classification, we compared the results to manually labeled data. This process was similar to the one used for metaphor identification. To enhance the robustness of the output, we used three different prompts and applied a majority voting system to obtain the final result. The baseline prompts were designed to instruct the model to identify and classify conceptual metaphors in a given set of sentences, returning the results in a standardized format. This approach helped mitigate variability in

responses and ensure consistency. An example of the prompt can be found in the appendix.

	Prompt 1	**Prompt 2**	**Prompt 3**
Identify, extract and classify conceptual metaphors. Classify into categories provided.	✓	✓	✓
Definition of Conceptual Metaphors	✓	✓	✗
Example Categories (taken from Hu et al.)	✓	✗	✗
Examples metaphors / Counter-example	5 / 3	2 / 0	9 / 0

Table 2: Comparison of Prompts

A "majority vote" approach was employed to determine the final result, ensuring that at least two out of three GPT responses agreed on a metaphor for it to be selected. This method involved connecting to the GPT-4o model with a temperature setting of 0 and a seed of 123. Each interview was processed individually, resulting in a total of 204 API calls (68 interviews x 3 prompts).

The total number of sentences sent for analysis was 577. Although ChatGPT occasionally returned sentences with slight grammatical variations, we used approximate string matching techniques to align the results with the initial sentences. Two methods were employed: Levenshtein distance [10] and Fuzzy Matching [5]. Fuzzy Matching proved to be more effective and appropriate for our needs, providing a higher degree of accuracy in matching the results to the initial sentences hence the following results are the ones from Fuzzy matching.

Given that this exercise proved to be accurate, we decided to label the entire dataset of ECB communications, thus significantly enhancing the efficiency and accuracy of our metaphor analysis.

The results presented in table 3, from the majority vote after the three GPT indicate notable consistency and slight variations in performance. We compare the count of identified metaphors per interview. Prompt 1 and Prompt 2 both returned 497 metaphors, achieving a ratio of 86.1%. Prompts 1 correctly identified 90.7% of the sentences and 80.4% of the

categories, while Prompt 2 identified 91.1% of the sentences but slightly lower at 78.1% for categories. Prompt 3 returned the highest number of metaphors at 505, with a ratio of 87.5%, and identified 91.1% of the sentences but had the lowest category identification rate at 77.8%. The majority vote method showed a slight improvement, resulting in 506 metaphors with the highest ratio of 87.7%, maintaining a sentence identification rate of 91.1% and achieving 80.0% for categories. These findings demonstrate the robustness and reliability of using a majority vote approach to enhance the accuracy of metaphor identification and categorization from our initial labels.

	Returned	**Ratio (%)**	**Sentences Identified (%)**	**Categories Identified (%)**
GPT 1	497	86.1	90.7	80.4
GPT 2	497	86.1	91.1	78.1
GPT 3	505	87.5	91.1	77.8
Majority Vote	506	87.7	91.1	80.0
Human Label	493	85.4	100	100

Table 3: Results Comparison of the 3 different prompts and the Majority vote decision

Out of the 505 categories, 7 were mistakenly categorized into different categories and thus added to 'other'. The lower categorization accuracy could be due to interpretation issues. Maintaining consistency in labeling has been challenging due to the subtle differences and context-dependent nature of many similar metaphors. One key difference noted between our manual labels and GPT's output is that the API classifies a considerable proportion of the observations into the 'other' category. In contrast, when reading the metaphors, we tended to assign them to one of our nine categories rather than 'other'.

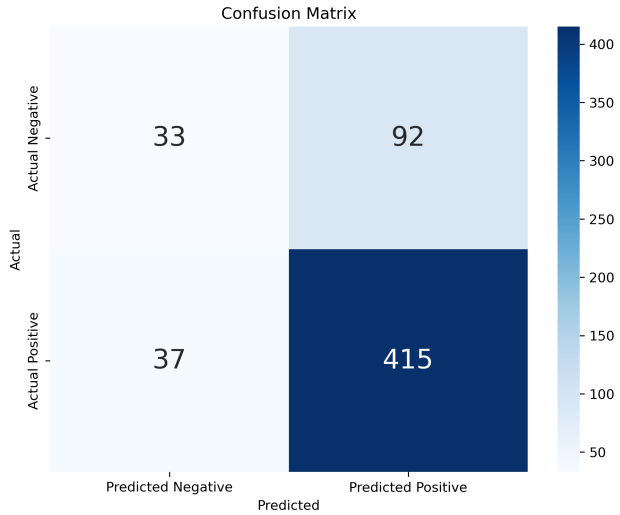

Figure 2: Confusion Matrix of Majority vote

3.4 Final Labels

Ultimately, GPT-4o was used to label the entire dataset, employing the same three prompts and retaining only the majority vote decisions. This involved 3 times 519 API calls for the 519 interviews. Out of the 3437 sentences we successfully matched 97.5% of the metaphors back to the initial sentences using Fuzzy matching. The majority vote approach resulted in 3083 metaphors, with 88.4% of the sentences being identified as metaphors

	Returned	**Ratio (%)**
GPT 1	3060	89.0
GPT 2	3073	89.4
GPT 3	2999	87.2
Majority Vote	3083	88.4

Table 4: Results Comparison

By using GPT-4o and the majority vote method, we achieved a high level of accuracy in identifying and categorizing metaphors, thereby demonstrating the effectiveness of AI-driven text analysis in linguistic studies. Given the success of this approach in our initial analysis, we will employ it to label the entire dataset of ECB communications. This will not only streamline the process but also enhance the accuracy and depth of our metaphorical analysis, contributing valuable insights to the field of AI-driven text analysis.

4 Metaphor Detection

This section represents the core of our study, where we progressively build more complex model architectures to detect metaphors. We begin with a simple word matcher using regular expressions, progress to a naive part-of-speech (POS) comparison, and ultimately apply a neural network model as described in Rei et al.[21] to capture deeper and more nuanced embeddings in the text.

For all the following methods, we employ a standard preprocessing pipeline. This involves lowercasing all words, removing special characters, numbers, and stopwords [3], and finally tokenizing the words for modeling. It is important to mention that we chose not to modify punctuation, as we need to preserve full stops to separate sentences.

To evaluate the performance of our models, we compare their detection with our 'augmented' manually labeled dataset. By using the GPT-labeled output as the true labels, we can subsequently compute performance metrics for our models.

4.1 Regular Expressions (REGEX)

Our first model is based on regular expressions to identify metaphors in the text. This method involves scanning sentences for the presence of both inflation-related terms and a set of predefined keywords. The list of relevant words for the metaphors is derived from the Hu et al. paper [11]. Despite its simplicity, this method mirrors the approach used in the paper mentioned previously, where metaphors are defined and subsequently categorized based on the presence of specific keywords. This approach aligns with the methodology we used for manually labeling the interviews and will be the closest to what is done in the paper.

In addition to the preprocessing pipeline mentioned earlier, we also apply lemmatization [17] to both our data corpus and our set of keywords. The logic behind this pruning approach is to account for variations of our words of interest that may appear in the text, ensuring we do not miss any relevant terms. For instance, the word *accelerate* may appear in the corpus as *accelerating*, and with the lemmatization method, we can match both variations to the dictionary form, lemma: *accelerate*.

4.2 Part of Speech (PoS)

Specifically, we identify relationships between the word "inflation" and a predefined set of keywords within three syntactic structures:

- Subject - Verb
- Verb - Object
- Adjective - Noun

POS tagging involves labeling each word in a sentence with its corresponding part of speech, such as a noun, verb, adjective, etc. This process helps in understanding the grammatical structure of the text. In our context, we use POS tagging to identify specific syntactic relationships that involve the word "inflation" and our set of keywords.

This method extends our initial Regex identifier, for a more precise prediction of metaphors. By checking for specific syntactic relationships, we can better determine the metaphoricity of matches. Naturally, we expect this model to predict fewer metaphors overall, but those identified should be more accurate representations of true metaphors.

Unlike the first model, here we are required to retain all tokens (words) in their original form initially. This is because Part of Speech (POS) tagging relies on analyzing words within the context of their sentences. Pruning the words to their base forms right away would strip away these important contextual interactions. Therefore, lemmatization must be performed after POS tagging and before matching keywords, ensuring we preserve the necessary context for accurate tagging, but we are also able to match out keywords as done in the previous model.

4.3 Neural Network

To enhance the detection and classification of metaphors, we employed a Neural Network approach. The rationale behind using a neural network is supported by the findings in the paper *Neural Metaphor Detection in Context* [9], which demonstrates their effectiveness in capturing the nuanced, metaphorical meaning of sentences. Given the task's complexity in understanding metaphorical context within sentences, neural networks provide a sophisticated method of analysis compared to simpler models.
We opted for a basic yet robust implementation, acknowledging that while our approach may not compete with the most advanced neural network models, it remains effective for the scope of this study. The dataset was split into an 80/20% training-test ratio, resulting in 2,749 sentences for training and 688 for testing.

4.3.1 Model Architecture

The model was designed for simplicity and efficiency, consisting of an embedding layer followed by dense layers with dropout and layer normalization to prevent overfitting. Specifically, we limited the architecture to a maximum of 3 hidden layers and a total

number of weights between 200,000 and 400,000. Given our relatively small data corpus, fitting a highly complex model may not be the most suitable choice for our problem, and we conducted an optimization search to find the best configuration.

4.4 Optimization

To improve the model, a grid search was conducted to explore different configurations of layers, learning rates, and other hyperparameters. This approach ensured the model achieved the best possible performance given the constraints. The optimal combination turned out to be two hidden layers with 220 nodes each, using the sigmoid activation function and a dropout rate of 0.5.

4.4.1 Model Summary

The final neural network model consisted of approximately 231,000 parameters, balanced between embedding and dense layers. This architecture, though basic, is effective in capturing the metaphorical meaning of sentences and provides a foundation for further refinement.

4.5 Pre-trained Large Language Model for Metaphor Detection

The decision to use a pre-trained Large Language Model (LLM) stems from our objective to assess the performance enhancement it offers without the computational overhead of training from scratch. In text classification, pre-trained large language models have demonstrated significant effectiveness, as evidenced by their successful application in various tasks.

Given the complexity of metaphor identification and the absence of available pre-trained models for metaphor identification or classification, we selected the general-purpose pre-trained model *distilbert-base-uncased* [22]. This model, though not specifically trained on metaphors, leverages its deep contextual understanding to perform effectively in this domain. We plan to fine-tune specific parameters of this transferred model to tailor it for our specific task setup.

As with our previous approaches, the dataset was divided into a train-test split of 80/20%, resulting in a training set of 2,749 sentences and a test set of 688 sentences.

4.5.1 Model Architecture

The model utilized the distilbert-base-uncased checkpoint, which consists of approximately 66.3 million parameters. This architecture begins with an embedding layer sourced from the DistilBERT model, followed by a dense layer to process the embeddings. To mitigate the risk of overfitting, a dropout layer is included in the design.

To effectively leverage the pre-trained embeddings of DistilBERT, the model's layers were initially frozen. This approach allows the dense layers to benefit from the robust pre-trained representations without modification, thus expediting the training process. Following a preliminary warm-up period, the layers were unfrozen to fine-tune the model for our specific task.

Hence, the training process was executed in two distinct phases:

- Warm-up Phase: The model was trained with the BERT parameters frozen for 10 epochs, allowing the dense layers to adapt and learn from the pre-trained embeddings without altering them.
- Fine-tuning Phase: The entire model, including the BERT layers, was then trained for an additional 20 epochs to fine-tune the parameters specifically for metaphor detection.

By integrating a pre-trained large language model, we harnessed the capabilities of advanced transformer-based architectures. This approach facilitated a more nuanced and precise understanding of metaphorical language, thereby making a significant contribution to the field of AI-driven text analysis.

5 Results

5.1 Metaphors Identification

In this section, we present the results from the identification models. These results were obtained by evaluating data from the test set, which consists of a total of 688 sentences. Our first finding is that the Regex model shows fairly consistent performance across all computed metrics, with values around 68%. The Part of Speech model, on the other hand, performs poorly overall. This can be attributed to the method's reliance on a specific word pair structure, leading to the prediction of very few metaphors. Finally, both the simple neural network (NN) and the imported Large Language model perform similarly, with the latter having an edge in all categories. This indicates that the LLM was able to capture

more nuanced embeddings in the text. However, when compared to the Regex model, the LLM cannot be definitively chosen over it. While the LLM achieves better accuracy, its recall is worse, indicating it does not offer a more comprehensive approach. This outcome is not surprising, as the Regex approach closely aligns with the methodology defined in the paper, upon which the metaphors' definitions are based.

	Regex (%)	**Part of Speech (%)**	**Neural Network (%)**	**LLM (%)**
Accuracy:	68.79	30.05	84.88	86.19
Precision:	68.10	43.80	60.53	67.67
Recall:	68.79	30.06	54.51	58.94
f1:	67.53	35.30	55.21	60.99

Table 5: Performance Comparison

Next, we plot in Figure 3 the distribution of the identification results for our four models and the GPT labels.

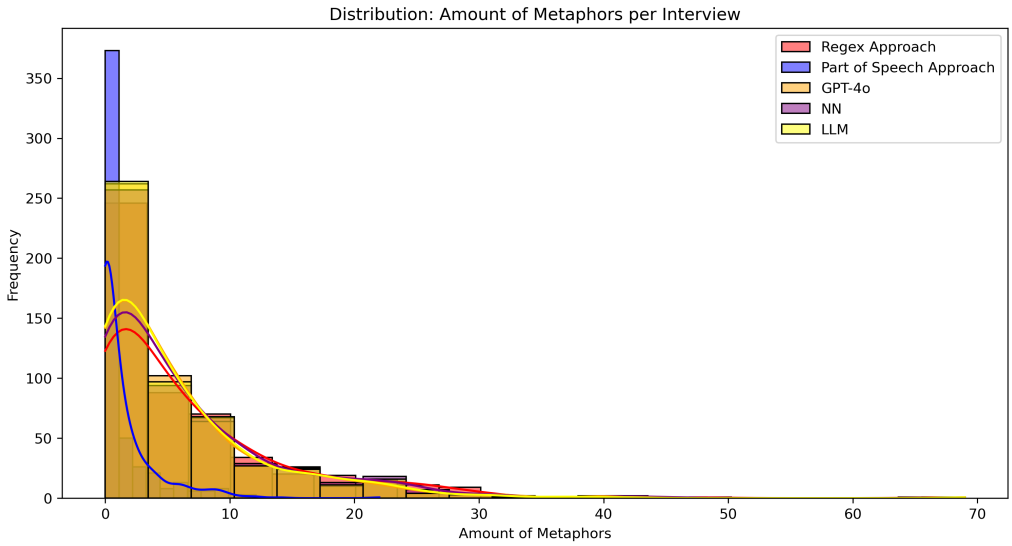

Figure 3: Distribution of Metaphors per Interview

We observe that, in terms of metaphor identification, the empirical distribution of the results closely mirrors the true labels, except for the Part of Speech model. As previously mentioned, the Part of Speech model captures fewer metaphors.

5.2 Descriptive Analysis of Metaphors

After having identified the metaphors related to inflation in ECB communications, we proceed by analyzing who employed these metaphors and the context in which they were used by ECB members. This analysis seeks to offer a more profound comprehension of the role and use of metaphors about inflation within ECB communications.

5.2.1 Analysis of Metaphor Usage by ECB Members

To determine which ECB executive board members use metaphors most frequently in their communications, we calculated the total number of words spoken by each of the 19 members throughout the entire period. Next, we quantified the total number of metaphors each member used during this period. By dividing the number of metaphors by the total number of words spoken by each member, we obtained a ratio representing the frequency of metaphor usage per word spoken. This ratio allowed us to compare the metaphor usage across members, independent of their overall volume of speech. The ratios of metaphors per word by member are plotted in figure 4.

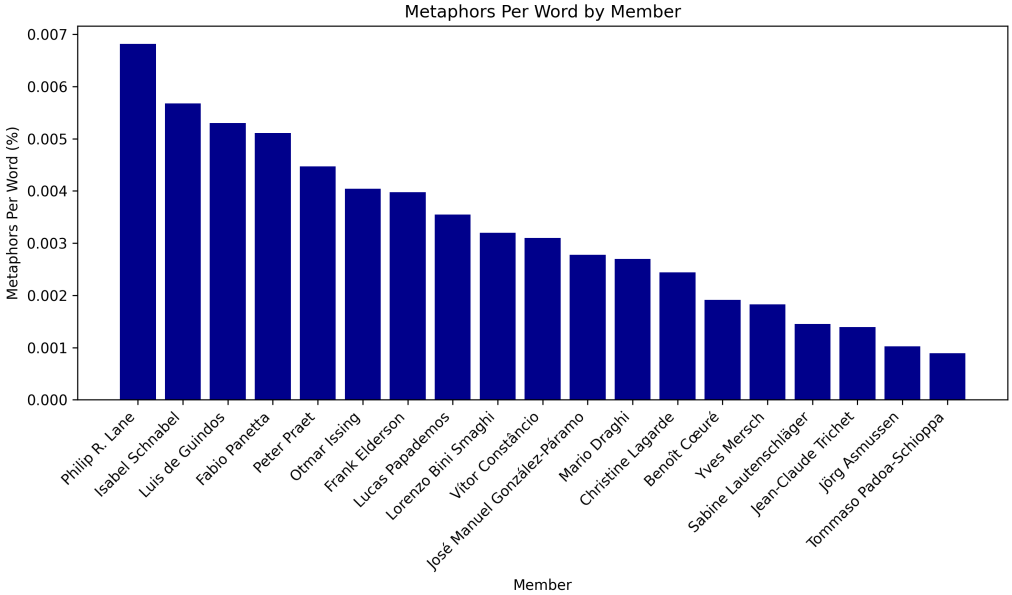

Figure 4: Ratio of Metaphors per Words Spoken by Member

Figure 4 shows that Philip R. Lane (Chief Economist and Executive Member at the ECB), Isabel Schnabel (ECB Executive Member), and Luis de Guindos (Vice-President and Executive Member at the ECB) are the top three members in terms of metaphor usage per word. Philip R. Lane employs an average of 7 metaphors per 1,000 words, Isabel Schnabel uses 6 metaphors per 1,000 words, and Luis de Guindos averages 5.5 metaphors per 1,000 words.

Subsequently, we studied metaphor usage by gender. We calculated the ratio of total metaphors used by gender to the total number of words spoken by gender. The results are presented in Figure 5.

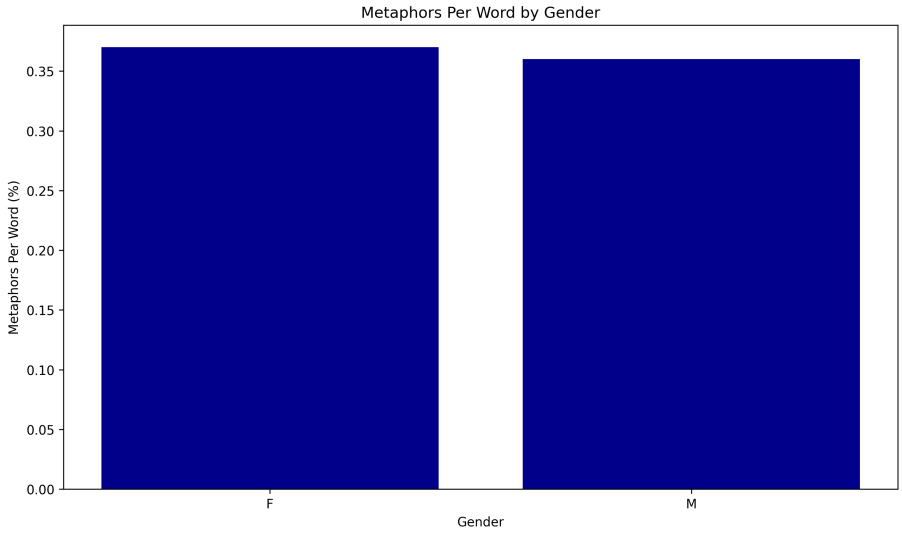

Figure 5: Metaphor Usage Ratios per Words Spoken by Gender

We observe no significant differences in metaphor about inflation usage between male and female ECB members.

Finally, we examined metaphor usage based on the media outlet conducting the interview with the ECB member. We calculate the ratio of metaphors per interview for the twenty media outlets with the highest number of interviews in the dataset. The results are depicted in Figure 6.

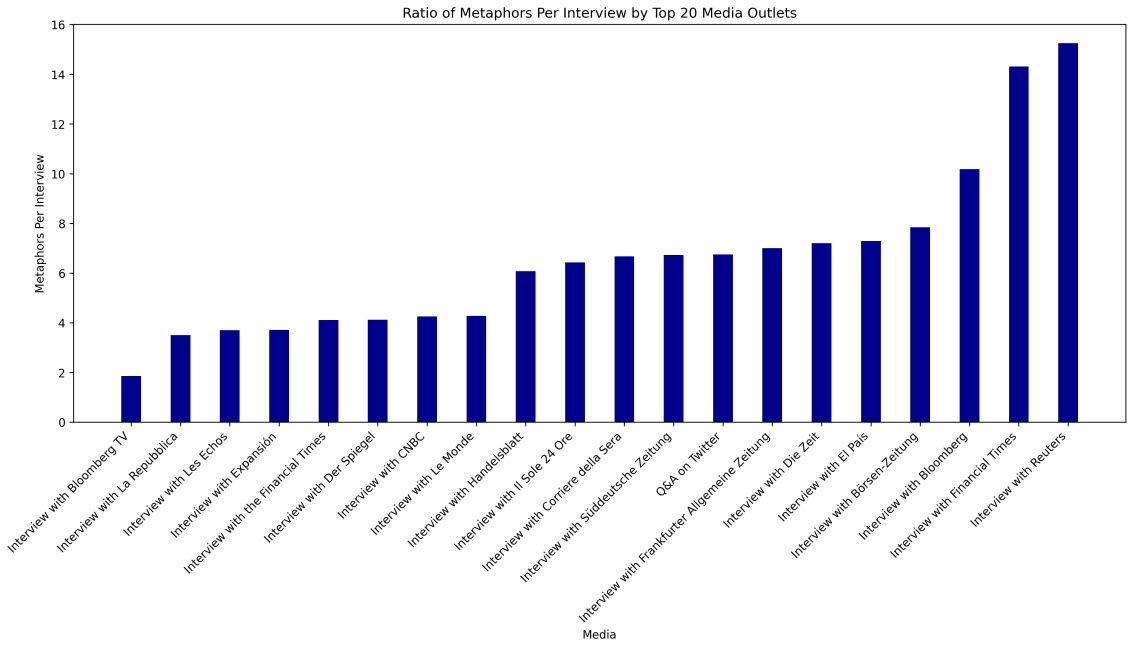

Figure 6: Metaphor Usage Ratios per Interview Across the Top 20 Media Outlets

Figure 6 shows that ECB members tend to use more metaphors when interviewed by Reuters, the Financial Times, and Bloomberg. On average, ECB members use 15.25 metaphors per interview with Reuters, 14.30 metaphors with the Financial Times, and 10.18 metaphors with Bloomberg. This suggests that ECB members may tailor their communication style based on the media outlet.

To further investigate this assumption, we categorized each media outlet manually into two groups. Outlets focusing on economics, business, and finance were classified as "Economics and Finance," while those with different focuses were classified as "General." Figure 7 plots the ratio of metaphors per interview for these two categories.

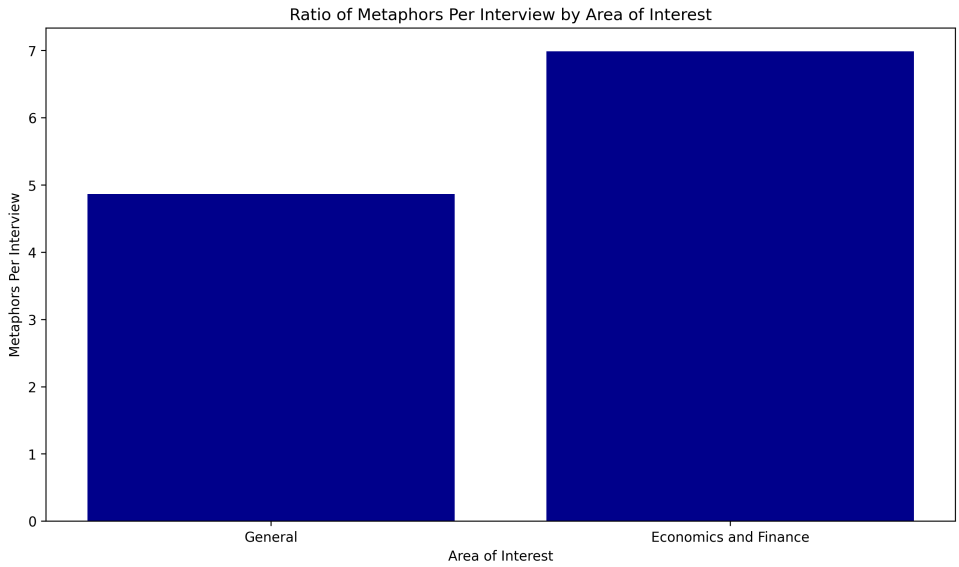

Figure 7: Metaphor Usage Ratios per Interview by Media Specialization

Figure 7 indicates that ECB members tend to use more metaphors per interview when speaking to media specializing in economics and finance. Specifically, ECB members employ an average of 7 metaphors per interview with economic and financial media, compared to 4.9 metaphors per interview with general media.

5.2.2 Analysis of Metaphor Categories

Next, we explore and try to understand which types of metaphors are most commonly used and whether any patterns exist concerning ECB members, gender, and media.

We seek to explore how the usage of each metaphor category varies across ECB members, genders, and media outlets. As a reminder, each metaphor was initially classified into one of 10 different categories: fire, disease, plant, warfare, orientation, sports, machine,

liquids, animal, and others. For example, the metaphor *"cool down inflation"* was classified under machine. The metaphors *"Even 3% inflation will hurt fixed payments eventually," "The battle against inflation is not over yet,"* and *"the threat of deflation could come to the fore"* were classified under warfare. Metaphors such as *"inflation tends to rise, " "maintaining inflation rates below but close to 2 percent,"* and *"inflation will converge,"* which illustrate ideas and concepts using directions, were classified under orientation. We then studied which metaphor categories were most commonly employed in ECB member speeches, as shown in Table 6. We decided to remove the "other" category, which accounts for 26% of all metaphors, as we believe it does not provide valuable insights into the ECB's communication style.

Category	**Count**	**Percentage (%)**
machine	238	10.610789
disease	279	12.438698
plant	66	2.942488
warfare	212	9.451627
fire	40	1.783326
orientation	1326	59.117254
liquids	39	1.738743
animal	18	0.802497
sports	25	1.114579

Table 6: Counts and Percentages of Metaphor Categories used by ECB Members

The most commonly used metaphor categories by ECB members are orientation, disease, machine, and warfare, accounting for 59.12%, 12.44%, 10.61%, and 9.45% of all identified metaphors, respectively. Before delving deeper into the usage of different metaphor categories, we chose to eliminate the "orientation" category. Metaphors under "orientation" were the most frequent, with terms like "higher inflation," "increasing inflation," and "the ability of inflation to return" often cited as conceptual metaphors in literature. However, we found that these terms do not provide valuable insights into the ECB's communication strategy. Instead, they align more closely with standard economic terminology rather than offering a distinct analogy about inflation.

Subsequently, we analyzed the types of metaphors about inflation used by each ECB executive member. Table 7 displays the most frequently used metaphor category for each ECB member, highlighting the highest counts in each row. For a more detailed breakdown, Table 9 in the appendix includes the number of metaphors used by category for each ECB member, with the highest counts in each row in bold.

Member	Most Frequent Category	Count
Benoît Cœuré	disease	29
Christine Lagarde	machine	18
Fabio Panetta	disease	13
Frank Elderson	disease	5
Isabel Schnabel	warfare	47
Jean-Claude Trichet	machine	29
José Manuel González-Páramo	disease	2
Jörg Asmussen	machine	4
Lorenzo Bini Smaghi	disease	1
Lucas Papademos	fire	4
Luis de Guindos	machine	41
Mario Draghi	disease	15
Otmar Issing	machine	1
Peter Praet	disease	38
Philip R. Lane	machine	53
Sabine Lautenschläger	disease	4
Tommaso Padoa-Schioppa	fire	1
Vítor Constâncio	disease	13
Yves Mersch	disease	7

Table 7: Counts and Percentages of Metaphor Categories used by ECB Members

Based on table 9, we then plotted Figure 8, which illustrates the percentage distribution of metaphor categories used by ECB members. This visualization allows for clearer comparisons of the different types of metaphors employed by each member. The figure only includes the eight ECB members who used more than 40 metaphors over the entire period.

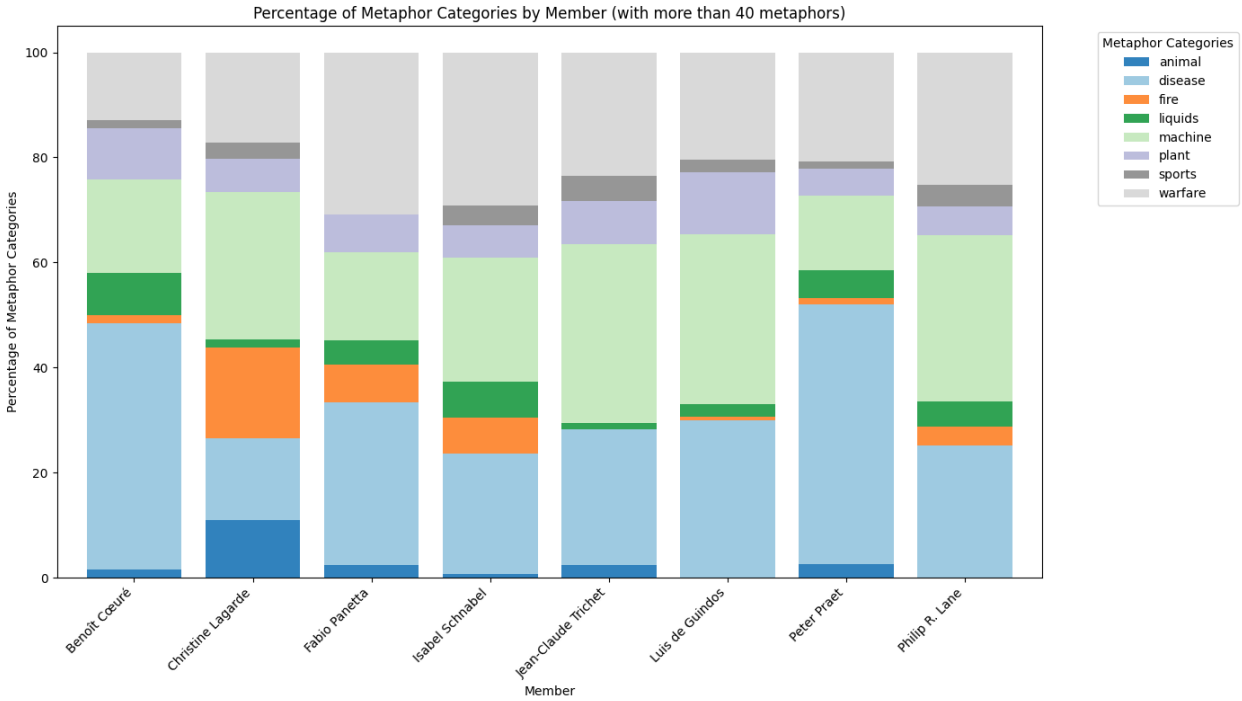

Figure 8: Percentage of Metaphor Categories Used by ECB Members with Over 40 Metaphors in Total

Figure 8 provides several interesting insights. Firstly, Christine Lagarde (President of the European Central Bank) uses significantly more metaphors about inflation from the fire and animal categories compared to other members. Fabio Panetta and Isabel Schnabel also employ more metaphors from the fire category relative to other members. Conversely, most other members seldom use metaphors from the fire category. Benoît Cœuré and Peter Praet tend to favor metaphors from the disease category. Metaphors from the machine and warfare categories are used by all members, and their usage is relatively uniform across the board.

Next, we look at the differences in the use of metaphor categories by gender. Results are shown in percentages in figure 9.

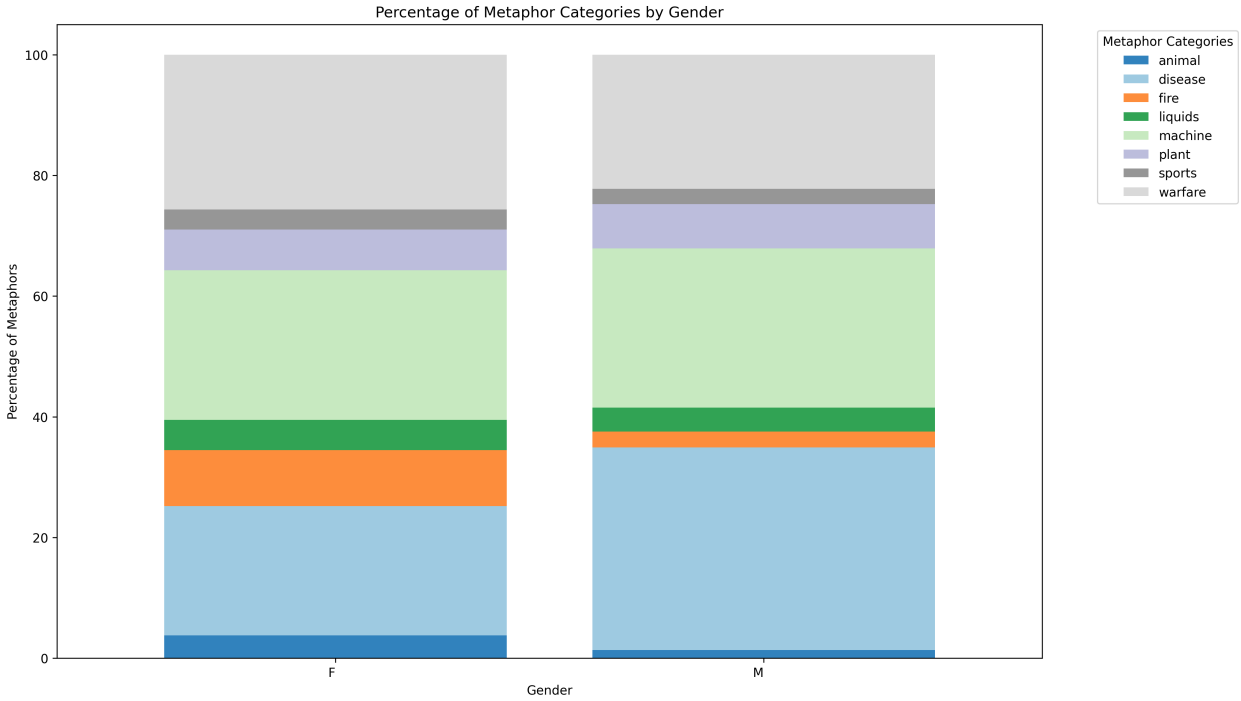

Figure 9: Percentage of Metaphor Categories Used by Each Gender

Figure 9 reveals several notable differences in the use of metaphor categories between male and female ECB executive board members. Female members tend to use more metaphors from the fire, animal, and warfare categories but fewer from the disease category compared to their male counterparts. Conversely, there are no significant differences between genders in the use of metaphors from the liquids, sports, machine, and plant categories.

Lastly, we investigate the relationship between the types of metaphors used by ECB members and the ten media outlets with the most interviews. First, we grouped all the metaphors by these top ten media outlets and summed the occurrences of each category (Table 10). To facilitate comparison, we converted these numbers into percentages. The results are illustrated in Figure 10.

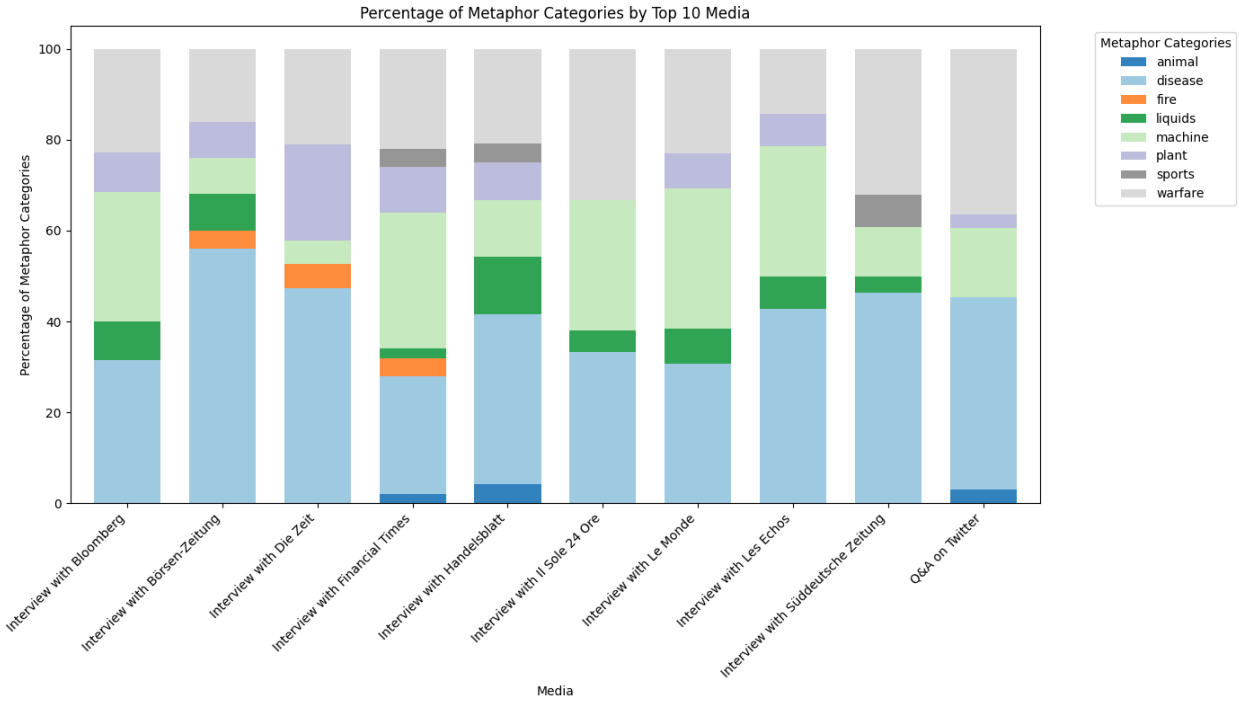

Figure 10: Percentage of Metaphor Categories Used by Each Gender

Next, we decide to examine the types of metaphors about inflation used based on the media's area of interest. Using the same media classification as before, we explored any differences in the types of metaphors employed by ECB members when interviewed by media specializing in economics and finance compared to general media. Our findings are presented in Figure 11.

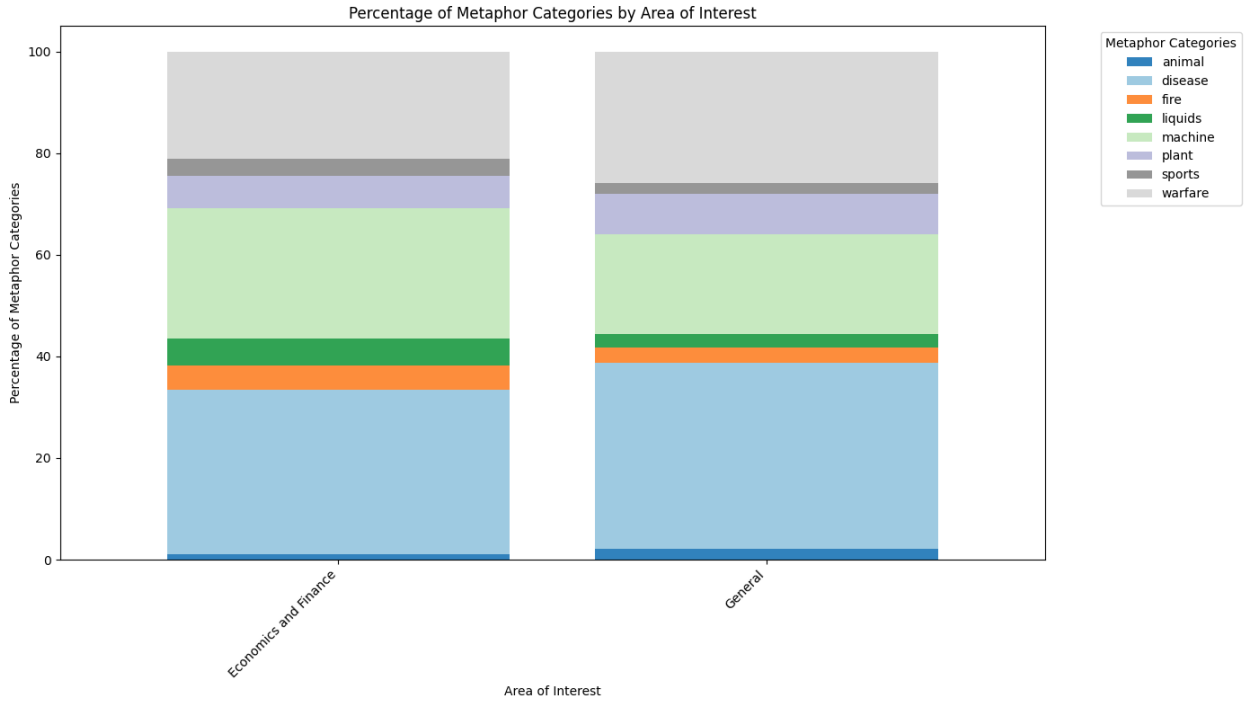

Figure 11: Percentage of Metaphor Categories by Media Specialization

Figure 11 shows distinct patterns in the metaphor usage by ECB members based on the type of media. Specifically, ECB members tend to employ more metaphors from the machine category and fewer from the sports and disease categories when interviewed by media specializing in economics and finance. Metaphors from other categories are used with similar frequency across both specialized and general media.

6 Conclusions

This thesis sets out to systematically identify and categorize conceptual metaphors related to inflation in the communications of ECB board members, employing advanced methodologies such as regular expressions, part-of-speech tagging, and neural networks. In particular, this study lays the foundation for further research.

Our analysis confirms that ECB board members consistently use metaphors when discussing inflation, reflecting the complex and abstract nature of economic concepts. There was no substantial improvement when implementing more complex models for detection. However, this result may stem from the strict definition of metaphors as per the referenced paper. Considering more flexibility in the structure of metaphors could potentially enhance the performance of large language models and lead to better detection improvements.

Among the identified metaphors, the most prevalent categories were orientational, warfare, and disease. These categories reveal the board members' tendencies to frame inflation in familiar and relatable terms, highlighting the underlying narratives that shape public understanding and policy perceptions. For instance, orientational metaphors suggest movement and directionality, warfare metaphors evoke a sense of struggle and conflict, and disease metaphors imply a pathological condition needing remedy. We also find that female executive members tend to use more metaphors from the fire, animal, and warfare categories compared to their male counterparts. Interestingly, more metaphors are employed when answering questions for economic-specific news portals compared to mainstream media. Contrary to our initial hypothesis that speakers might use more metaphors when addressing a broader audience, it appears that they employ more metaphors when communicating with more specialized listeners.

The use of the ChatGPT API was instrumental in automating the labeling and categorization process, ensuring efficiency and accuracy. This integration underscores the potential of leveraging advanced language models in linguistic research, offering new avenues for future studies.

This study provides an overview of metaphor usage by ECB board members and paves the way for further research that could delve into the specific functions of these metaphors and explore the distinctions among different metaphor categories. Additionally, examining how these metaphors influence public perception and policy effectiveness would be a valuable extension of this work.

In conclude, the findings affirm that metaphors are a pivotal element in the discourse on inflation among ECB board members, shaping both the communication strategy and the interpretative frameworks of economic policies.

A Appendix

Prompt 1:

Title: Conceptual Metaphor Detection

Description: Detect and extract conceptual Metaphors in a set of sentences.

Task: Given a set of sentences, you are required to identify and extract conceptual metaphors present. Only consider conceptual metaphors related to inflation, deflation, inflationary, disinflationary, hyperinflation, and disinflation. After identifying the metaphors, return the sentences that contain the metaphors and label it in the format:

Sentence: [Sentence text] Category: [Category]

Where, Category is one of the following 10 options (with a set of examples):

Fire = ['ignite', 'fuel', 'spark', 'dampen',...']

Liquids = ['erode', 'surge', 'subside', 'simmers',...]

Plant = ['hedge', 'nip']

Animal = ['soar', 'rampant', 'curb', 'tame', 'rein', 'creep', 'whip', 'gallop', 'halt', 'roar',...]

Disease = ['plague', 'worsen', 'benign', 'virulent',...]

Warfare = ['threat', 'subdue', 'target', 'beat', 'preempt', 'vigilant',...]

Sports = ['pace', 'outstrip', 'lag', 'surpass', 'quicken',...]

Machine = ['accelerate', 'faster', 'control', 'slow', 'trigger',...]

Orientation = ['low', 'rise', 'high', 'increase', 'reduce', 'decline', 'fall', 'exceed', 'spiral', 'hyper', 'upward', 'uptick',...]

Other = [any other metaphor not covered by the above categories]

Definition: Conceptual metaphors refer to linguistic expressions that metaphorically represent abstract concepts in terms of more concrete concepts.

Identifying Metaphorical and Non-Metaphorical Usage: To identify whether the use of "inflation" is metaphorical or non-metaphorical, consider the following questions:

Is another domain being used to describe inflation?: Check if the sentence draws on concepts from a different area of experience to explain or depict inflation. Metaphorical: "Inflation is eroding our savings." (Here, "eroding" draws on the domain of natural wear and tear, often associated with physical erosion.) Non-Metaphorical: "Inflation is measured by the Consumer Price Index." (This is a straightforward, literal description.)

Is the language used more concrete and vivid?: Metaphors often use vivid and concrete imagery to make abstract ideas more relatable. Metaphorical: "Inflation is a beast that must be tamed." (The use of "beast" provides a vivid and concrete image.) Non-Metaphorical: "Inflation affects the purchasing power of money." (This is a direct explanation without vivid imagery.)

Is the purpose to simplify and relate?: Metaphors are often used to simplify complex ideas

and relate them to common experiences. Metaphorical: "The economy is overheating due to inflation." (This uses the concept of overheating, which is a familiar experience, to explain economic stress.) Non-Metaphorical: "High inflation rates can lead to increased interest rates." (This is a factual statement explaining a cause-and-effect relationship.)
Additional Instructions: - If one sentence contains multiple metaphors, return the sentence multiple times with each metaphor labeled separately.
Example Input: ["China worries that such rapid growth could ignite inflation, and the countries banking industry has been trying since last year to bring the rate down to 7 percent a year.", "But the evidence from that earlier period is tainted by the fact that inflation was roaring at the time creating bracket creep, over-taxation of illusory capital gains and paper profits, and rising interest rates."]
Example Output: "Sentence: China worries that such rapid growth could ignite inflation, and the countries banking industry has been trying since last year to bring the rate down to 7 percent a year. Category: Fire", "Sentence: "But the evidence from that earlier period is tainted by the fact that inflation was roaring at the time creating bracket creep, over-taxation of illusory capital gains and paper profits, and rising interest rates." Category: Animal"
The output has to be in the right format as shown in the example output. The categories are predefined and you have to choose the right category based on the metaphor used in the sentence."

Metaphor Categories by Member:

Member	Animal	Disease	Fire	Liquids	Machine	Plant	Sports	Warfare	Total
Benoît Cœuré	1	**29**	1	5	11	6	1	8	62
Christine Lagarde	7	10	11	1	**18**	4	2	11	64
Fabio Panetta	1	13	3	2	7	3	0	**13**	42
Frank Elderson	2	**5**	0	0	2	1	0	4	14
Isabel Schnabel	1	37	11	11	38	10	6	**47**	161
Jean-Claude Trichet	2	22	0	1	**29**	7	4	20	85
José Manuel González-Páramo	0	**2**	0	1	0	0	0	1	4
Jörg Asmussen	0	0	0	0	**4**	0	0	0	4
Lorenzo Bini Smaghi	0	**1**	0	0	0	0	0	1	2
Lucas Papademos	0	3	**4**	1	0	1	0	0	9
Luis de Guindos	0	38	1	3	**41**	15	3	26	127
Mario Draghi	1	**15**	1	2	9	3	1	5	37
Otmar Issing	0	0	0	0	**1**	0	0	1	2
Peter Praet	2	38	1	4	11	4	1	**16**	77
Philip R. Lane	0	42	6	8	**53**	9	7	42	167
Sabine Lautenschläger	1	**4**	0	0	3	2	0	3	13
Tommaso Padoa-Schioppa	0	0	**1**	0	0	0	0	0	1
Vítor Constâncio	0	**13**	0	0	9	0	0	11	33
Yves Mersch	0	**7**	0	0	2	1	0	3	13
Total	18	279	40	39	238	66	25	212	917

Table 8: Counts and percentages of metaphor categories by member

Metaphor Categories by Media:

Media	Animal	Disease	Fire	Liquids	Machine	Plant	Sports	Warfare	Total
Interview with Bloomberg	0	11	0	3	10	3	0	8	35
Interview with Börsen-Zeitung	0	14	1	2	2	2	0	4	25
Interview with Die Zeit	0	9	1	0	1	4	0	4	19
Interview with Financial Times	1	13	2	1	15	5	2	11	50
Interview with Handelsblatt	1	9	0	3	3	2	1	5	24
Interview with Il Sole 24 Ore	0	7	0	1	6	0	0	7	21
Interview with Le Monde	0	4	0	1	4	1	0	3	13
Interview with Les Echos	0	6	0	1	4	1	0	2	14
Interview with Süddeutsche Zeitung	0	13	0	1	3	0	2	9	28
Q&A on Twitter	1	14	0	0	5	1	0	12	33
Total	3	100	4	13	53	19	5	65	262

Table 9: Counts and percentages of metaphor categories by top 10 medias

References

[1] Aghazadeh, E., Fayyaz, M., and Yaghoobzadeh, Y. (2022). Metaphors in pre-trained language models: Probing and generalization across datasets and languages. *arXiv preprint arXiv:2203.14139*.

[2] Alvarez, L. F., Boxho, S., and Breier, M. (2024). Dsdm thesis. https://github.com/lalvarezpoli/DSDM_Thesis.

[3] Bird, S., Klein, E., and Loper, E. (2023). Natural language toolkit: Stopwords corpus. https://www.nltk.org/nltk_data/.

[4] Charteris-Black, J. (2004). *Corpus Approaches to Critical Metaphor Analysis*. Palgrave Macmillan, New York.

[5] D., A. (2019). A fuzzy approach to approximate string matching for text retrieval in nlp. *Journal of Computational Information Systems*, 15:26–32.

[6] Den Hartog, D. N. and Verburg, R. M. (1997). Charisma and rhetoric: Communicative techniques of international business leaders. *The Leadership Quarterly*, 8(4):355–391.

[7] Do Dinh, E.-L. and Gurevych, I. (2016). Token-level metaphor detection using neural networks. In *Proceedings of the Fourth Workshop on Metaphor in NLP*, pages 28–33.

[8] European Central Bank (2024). Timeline of the ecb's executive board.

[9] Gao, G., Choi, E., Choi, Y., and Zettlemoyer, L. (2018). Neural metaphor detection in context. In Riloff, E., Chiang, D., Hockenmaier, J., and Tsujii, J., editors, *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 607–613, Brussels, Belgium. Association for Computational Linguistics.

[10] Haldar, R. and Mukhopadhyay, D. (2011). Levenshtein distance technique in dictionary lookup methods: An improved approach.

[11] Hu, C. and Chen, Z. (2019). Inflation metaphor in contemporary american english. *Journal of Pragmatics*, 146:72–84.

[12] Jang, H., Moon, S., Jo, Y., and Rose, C. (2015). Metaphor detection in discourse. In *Proceedings of the 16th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 384–392.

[13] Lakoff, G. and Johnson, M. (1980). *Metaphors We Live By*. University of Chicago Press, Chicago.

[14] Liu, J., O'Hara, N., Rubin, A., Draelos, R., and Rudin, C. (2020). Metaphor detection using contextual word embeddings from transformers. In *Proceedings of the Second Workshop on Figurative Language Processing*, pages 250–255.

[15] Marrugo, D., Martinez Santos, J. C., and Puertas, E. (2023). Natural language content evaluation system for multiclass detection of hate speech in tweets using transformers.

[16] Moritz, E. (manuscript). Chatting with chat(gpt-4): Quid est understanding?

[17] Müller, T., Cotterell, R., Fraser, A., and Schütze, H. (2015). Joint lemmatization and morphological tagging with lemming. In Màrquez, L., Callison-Burch, C., and Su, J., editors, *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 2268–2274, Lisbon, Portugal. Association for Computational Linguistics.

[18] Neidlein, A., Wiesenbach, P., and Markert, K. (2020). An analysis of language models for metaphor recognition. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 3722–3736.

[19] OpenAI (2024). Openai api. https://openai.com/index/openai-api/.

[20] Picard, L. and Stammbach, D. (2019). Political metaphors in u.s. governor speeches. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3658–3664.

[21] Rei, M., Bulat, L., Kiela, D., and Shutova, E. (2017). Grasping the finer point: A supervised similarity network for metaphor detection. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 1539–1548.

[22] Sanh, V., Debut, L., Chaumond, J., and Wolf, T. (2019). Distilbert: A distilled version of bert for fast inference.

[23] Schulder, M. and Hovy, E. (2014). Metaphor detection through term relevance. In *Proceedings of the Second Workshop on Metaphor in NLP*, pages 18–26.

[24] SeleniumHQ (2023). Seleniumhq website. https://github.com/SeleniumHQ/seleniumhq.github.io/tree/trunk.

[25] Shutova, E., Kiela, D., and Maillard, J. (2016). Black holes and white rabbits: Metaphor identification with visual features. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 160–170.

[26] Sun, S. and Xie, Z. (2018). Bilstm-based models for metaphor detection. In *Natural Language Processing and Chinese Computing: 6th CCF International Conference, NLPCC 2017, Dalian, China, November 8–12, 2017, Proceedings 6*, pages 431–442. Springer.

[27] Venturini, A. (2020). Analysing the impact of ecb communication on financial markets: A text mining approach.

[28] Wachowiak, L. and Gromann, D. (2023). Does GPT-3 grasp metaphors? identifying metaphor mappings with generative language models. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1018–1032, Toronto, Canada. Association for Computational Linguistics.

[29] Wu, C., Wu, F., Chen, Y., Wu, S., Yuan, Z., and Huang, Y. (2018). Neural metaphor detecting with cnn-lstm model. In *Proceedings of the workshop on figurative language processing*, pages 110–114.